Find the first wrong step and repair it

LEARNING TARGET I can use a picture to identify a mistake and explain a correction.

SECTION A KEY VOCABULARY

Quantity	Name what is found: position, speed, etc.
Sign	Direction uses signs; speed is a size.
Force	Sum all forces on the same object.
Check	Use a graph or another equation.

SECTION B KEY EQUATIONS + UNITS

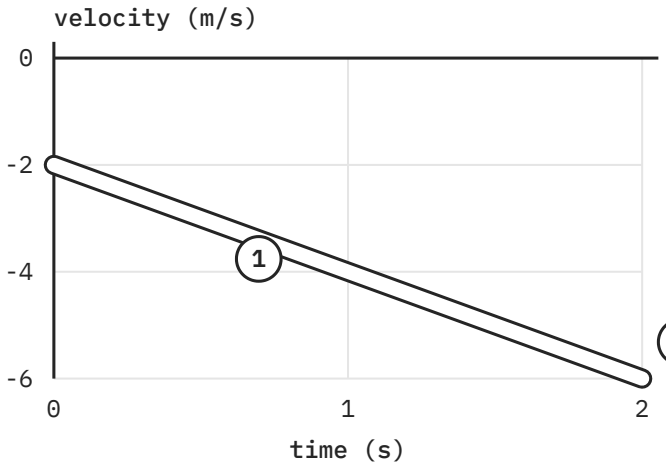
$$\Sigma F = ma$$

$$\Delta v = v_f - v_i$$

units N/kg = m/s²
repair first wrong step

SECTION C COLOR IT · CHECK THE CHANGE IN v

-2 to -6 m/s in 2 s.



~~Wrong: $\Delta v = +4$~~ 3

$\Delta v = -6 - (-2)$

$= -4$ m/s

2 speed: 2 → 6 m/s

☐ **1 · PURPLE** Velocity line

☐ **2 · GREEN** Increasing speed

☐ **3 · PENCIL** Incorrect Δv

SECTION D WORKED EXAMPLE

A **3 kg** cart has **9 N** right and **3 N** left. Someone writes $a = 9/3 = 3$ **m/s². Explain and repair the first wrong step.**

GIVEN	EQUATION	WORK	ANSWER
m = **3 kg** right is + forces: +**9 N** and −**3 N**	Use net force, not the largest force.	$\Sigma F_x = 9 - 3 = +\mathbf{6}$ **N** $a_x = 6/3 = +\mathbf{2}$ **m/s²**	+**2 m/s²** Check: ma = **6 N**.

SECTION E THE 2026-XJ THREAD

DAY 19 OF 22 · NEW TOOL: REASONING CHECKS

Before the fictional XJ forecast is shared, check its signs, units, starting coordinates and force sums. Repairing the first wrong step can change every later result.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else